

Antimicrobial agents used in treatment of infectious disease

Definition:

- **Antimicrobial agent:** A general term for drugs, chemicals, or other substances that either kill or slow the growth of microbes.
- Antimicrobial medicines can be grouped according to the microorganisms they act primarily against.
 - Antibiotics
 - Antifungals
- The use of antimicrobial medicines to treat infection is known as antimicrobial chemotherapy
- Antimicrobial prophylaxis antimicrobial medication to prevent disease

- Most microbiologist distinguish two groups of antimicrobial agents used in the treatment of infectious disease:
 - **Antibiotics**, which are natural substances of microorganisms, and
 - **Chemotherapeutic agents**, which are chemically synthesized.
- Also **semi synthetic and synthetic antibiotics** groups of antimicrobial agents used

☐ **Naturally occurring antimicrobials**

- Metabolic products of bacteria and fungi
- Reduce competition for nutrients and space
 - **Bacteria** – Streptomyces, Bacillus,
 - **Molds** – Penicillium, Cephalosporium
- Production in quantity is by **industrial fermentation**

Characteristics of Antibiotics

❑ A clinically-useful antibiotic should have as many of these characteristics as possible

- A wide spectrum of activity
- Nontoxic to the host and without undesirable side effects
- Soluble in body fluids,
- Nonallergenic to the host
- Not eliminate the normal flora of the host
- Able to reach the part of the human body where the infection is occurring.
- Inexpensive and easy to produce
- Chemically-stable (have a long shelf-life).
- Microbial resistance is uncommon and unlikely to develop.

- The range of bacteria or other microorganisms that are affected by a certain antibiotic is expressed as **spectrum of action**.
- Antibiotics effective against prokaryotes which kill or inhibit a wide range of **Gram-positive** and **Gram-negative** bacteria are said to be **broad spectrum**.
- If effective mainly against **Gram-positive** or **Gram-negative** bacteria, they are called **narrow spectrum**.
- If effective against a single organism or disease, they are referred to as **limited spectrum**.
- The deciding factor in the spectrum of antimicrobial activity is a **porin protein** on cell wall

Classification of Antibiotics

- One way to classify antibiotics is based on the targets in the microbial cell accordingly, there are six major classes
 - Cell wall synthesis inhibitor
 - Cell membrane disruptor
 - metabolites synthesis inhibitor
 - DNA synthesis (replication) inhibitor
 - RNA synthesis (transcription) inhibitor
 - Protein synthesis (translation) inhibitor
- Another way to classify antibiotics is based on their effect on growth and survival of the bacteria.
 - Bacteriostatic and
 - bactericidal antibiotics

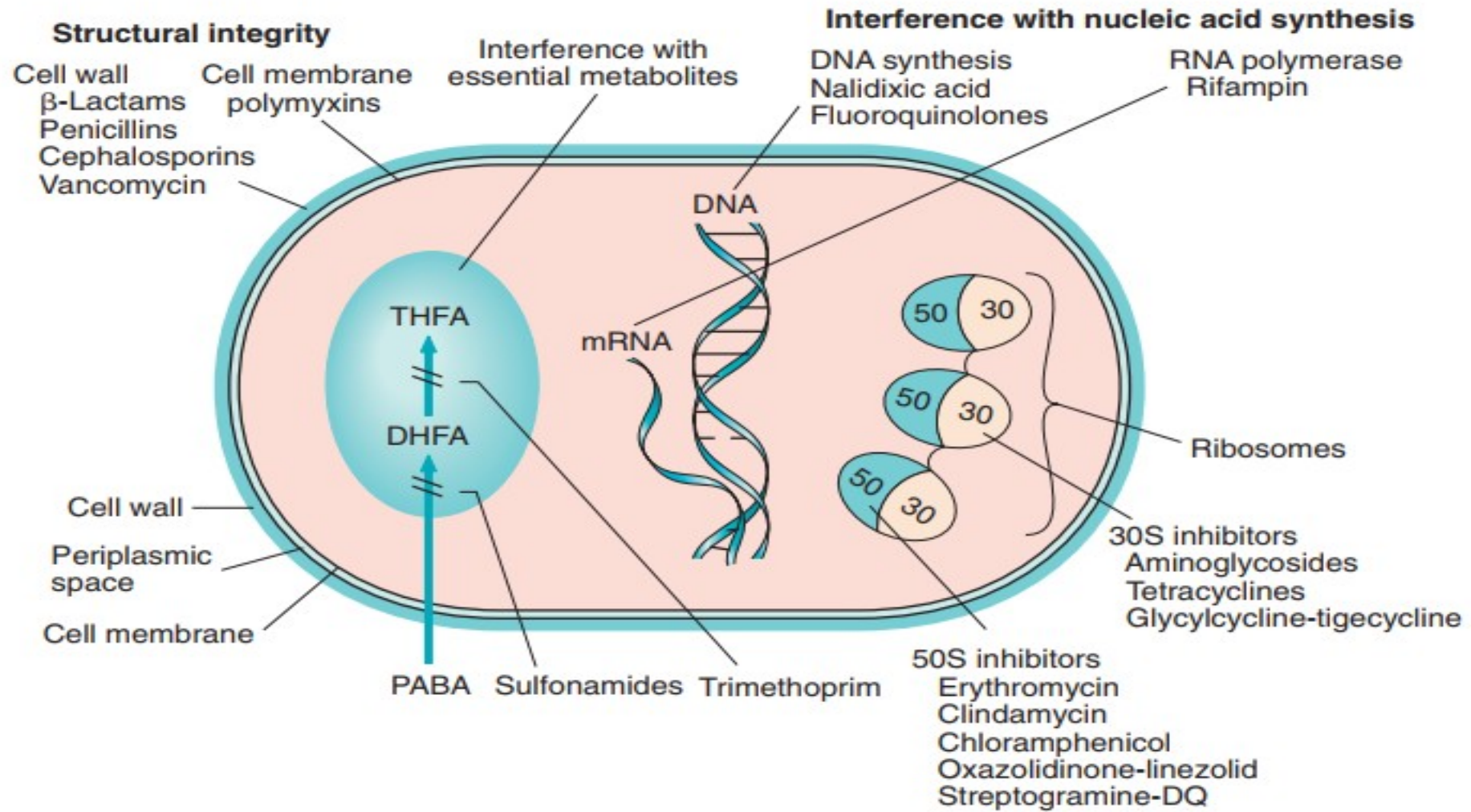
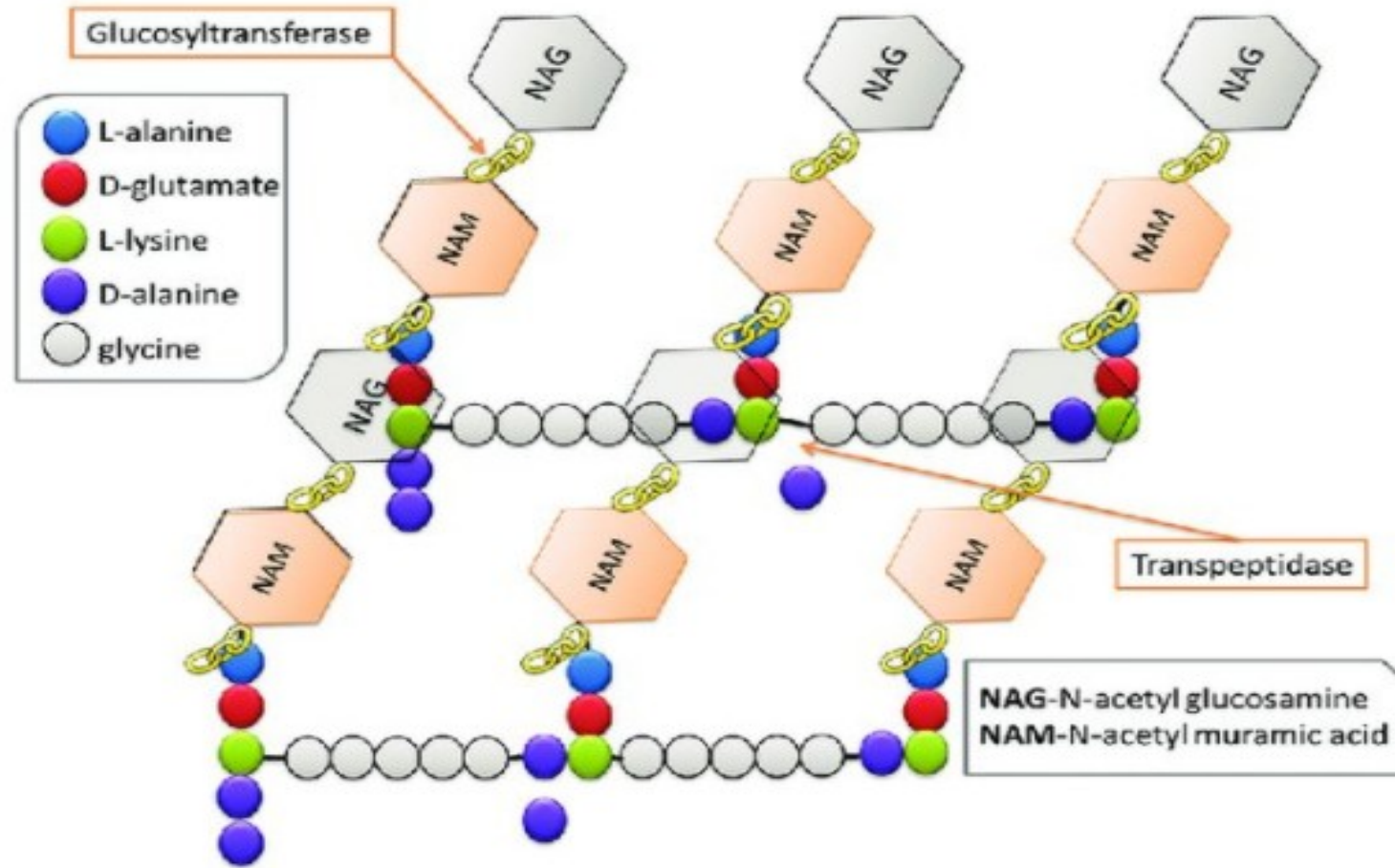


Fig1: Primary sites of antibacterial action for major classes of antimicrobial agents. *THFA*, Tetrahydrofolic acid; *DHFA*, dihydrofolic acid; *PABA*, para-aminobenzoic acid.

The effect of antibiotics against cell wall

- Cell walls of microorganisms are constructed by **peptidoglycan**.
 - **critical attack** site of anti-cell-wall agents
- This giant molecule is formed by weaving the linear glycans *N*-acetylglucosamine and *N*-acetylmuramic acid into a basket-like structure.
- Mature peptidoglycan is held together by **cross-linking** of short peptide side chains hanging off the long glycan molecules.
- This cross-linking process is the target of two of the most important groups of antimicrobics,
 - **Beta-lactams**
 - **Glycopeptides (vancomycin and teicoplanin)**

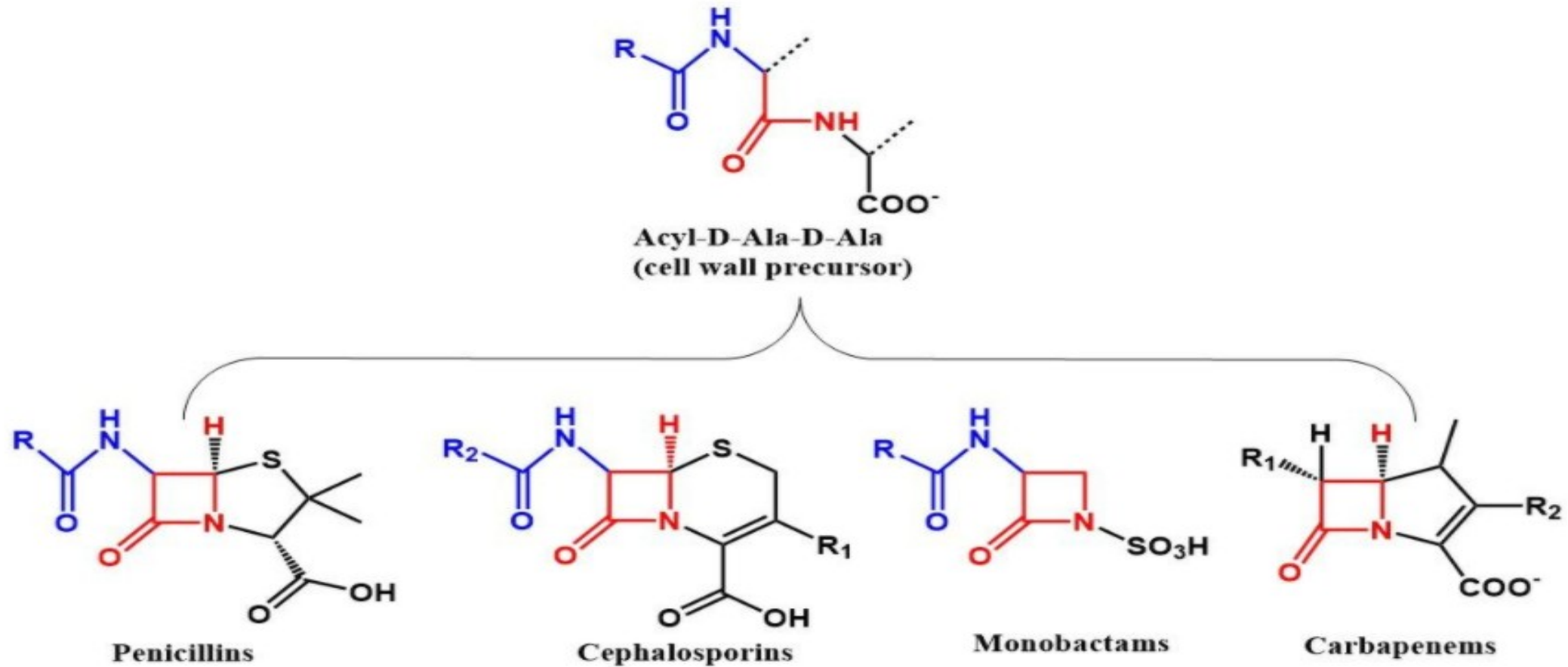
The structure of the peptidoglycan cell-wall



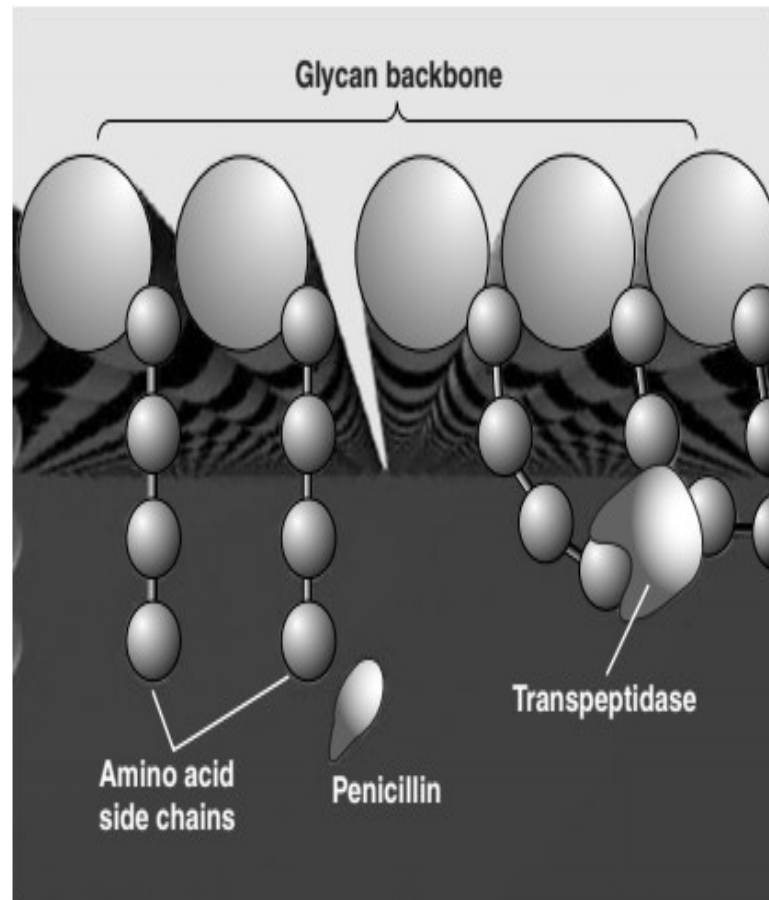
β-Lactam Antimicrobials

- Their name derives from the presence of a **β-lactam ring** in their structure; this ring is essential for antibacterial activity
- The β-Lactam antimicrobials comprise
 - ✓ penicillins, cephalosporins, carbapenems, and monobactams
- The β-lactam antimicrobials interfere with the **transpeptidation** reactions that seal the peptide crosslinks between glycan chains
- It leads to formation of bacteria with weak cell wall and undergo lysis
- These targets of all the β-lactams are commonly called penicillin-binding proteins (PBPs)
- Several distinct PBPs occur in any one strain, are usually species specific, and vary in the avidity of their binding to different β-lactam antimicrobics.

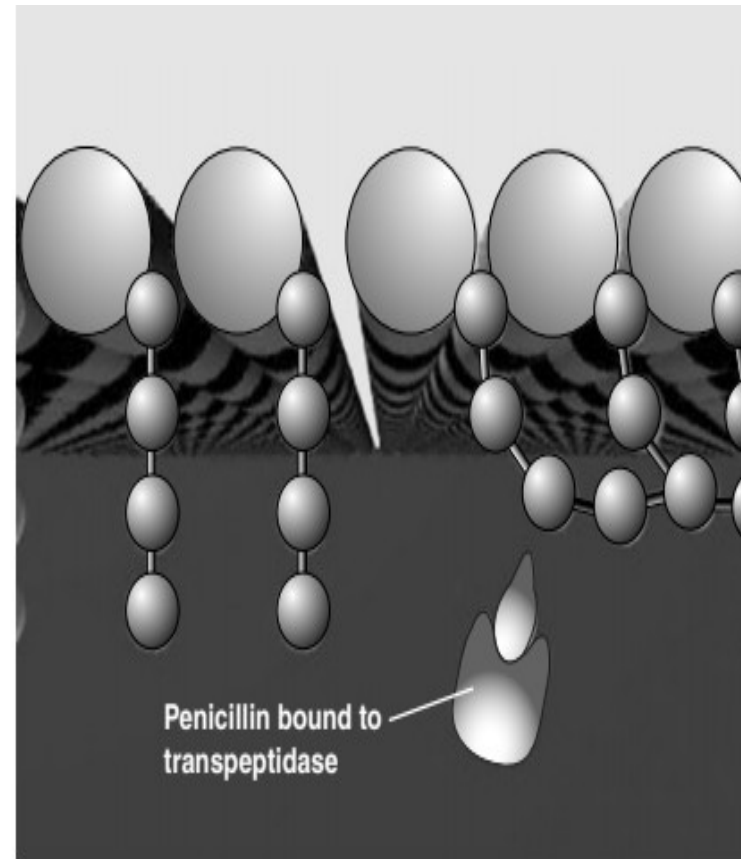
Structure and classification of beta-lactam antibiotics



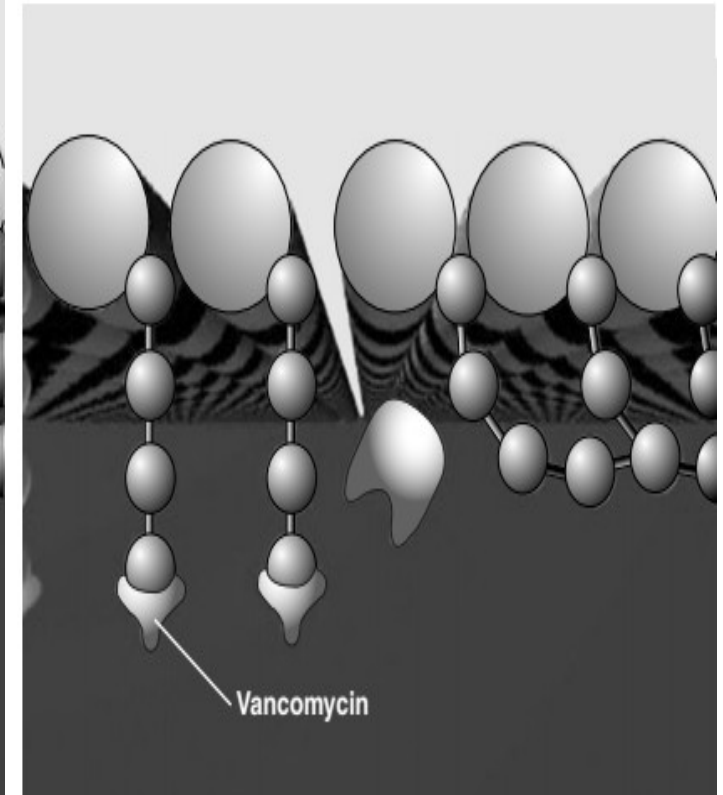
Core structure of beta-lactam antibiotics



A



B



C

Fig.2: Action of antimicrobials on peptidoglycan synthesis

- β -Lactam antimicrobials are usually highly bactericidal, but only to growing bacteria synthesizing new cell walls.
- Killing involves
 - Attenuation and disruption
 - Liberation or activation of autolytic enzymes that further disrupt weakened areas of the wall, and
 - Finally osmotic lysis from passage of water through the cytoplasmic membrane to the hypertonic interior of the cell

Clinical Use

- Usually the **drugs of choice** for infections by **susceptible organisms** because of their **low toxicity** and **bactericidal action**.
- They have also proved of great value in the prophylaxis of many infections.
- Penicillin reach the **CSF** when the meninges are inflamed and are effective in the treatment of **meningitis**, but 1st and 2nd GCs are not.
- In contrast, the 3rd GCs penetrate much better and have become the agents of choice in the treatment of undiagnosed meningitis and meningitis caused by most Gram-negative organisms.

Glycopeptide Antimicrobials

- Two agents, **vancomycin** and **teicoplanin**
- Vancomycin interrupts cell wall synthesis by forming a complex with the **C-terminal** **Dalanine** residues of peptidoglycan precursors.
 - prevents the transfer of the precursors from a lipid carrier to the growing peptidoglycan wall by **transglycosidases**.
 - Biochemical reactions in the cell wall catalyzed by transpeptidases and D,D-carboxypeptidases are also inhibited by vancomycin and other glycopeptide antimicrobials.
- The effect is the same as with β -lactams, prevention of peptidoglycan cross-linking; are **bactericidal**, but are primarily active only against Gram-positive bacteria.
- Their main use has been against **multi-resistant** Gram-positive infections including those caused by strains of staphylococci that are resistant to the penicillinase-resistant penicillins and cephalosporins.

Antibiotics that Affect the Function of Cytoplasmic Membranes

- Biologic membranes are composed basically of lipid, protein, and lipoprotein.
- The cytoplasmic membrane acts as a **diffusion barrier** for water, ions, nutrients, and transport systems.
- Cytoplasmic content and gradient such as micro/macromolecules and ions are controlled by active transportation via **integral transporter proteins**.
- When selective membrane permeability is disrupted by antimicrobials, ions are lost and cellular ion gradient is distorted, so, the organism undergoes cellular damage and death
- Targets of antimicrobials are metabolic steps of fatty acid synthesis and membrane phospholipids

- **Polymyxin B** and **Colistin** have a cationic detergent-like effect.
- Bind to the cell membranes of susceptible Gram-negative bacteria and alter their permeability, resulting in loss of essential cytoplasmic components and bacterial death.
- Their spectrum is essentially Gram-negative; they act against *P. aeruginosa* and other Gram-negative rods.
- These agents react to a lesser extent with cell membranes of the host, resulting in nephrotoxicity and neurotoxicity.
- The polymyxins are of virtually no use as systemic agents since they bind to various ligands in body tissues and are potent toxins for the kidney and nervous system.

Antibiotics That Inhibit Protein Synthesis



- Inhibitors of RNA synthesis will also have a secondary effect on protein synthesis.
- Each of the steps in the process of protein synthesis can be potential targets for antibiotics development
- An error in the sequence of one protein molecule has no deleterious effect because there are multiple other copies of the protein molecule that will have the right sequence.
 - antibiotics should be able to cause changes in the protein sequence every time the ribosome synthesizes
 - which requires tight binding of the antibiotic to the ribosome.
- Most inhibitors of protein synthesis are **bacteriostatic** so proper protein synthesis can resume once the antibiotic is removed.
- Thus the effect of these antibiotics is to prevent growth of the bacteria.

❑ Antibiotics can inhibit protein synthesis by targeting either

➤ 30S subunit ribosome

- tetracycline , and the **aminoglycosides**
- blocking of the ribosome
- misreading of the genetic code
- membrane damage
- irreversible uptake of the antibiotics.

➤ 50S subunit ribosome

- clindamycin, chloramphenicol
- blocking the action of peptidyl transferase
- blocks the path by which growing peptides exit the ribosome

Protein Synthesis inhibitor Antibiotics with Unusual Mechanisms of Action

- **Thermorubin**: binds to the **full 70S** ribosome inhibits initiation stage of protein synthesis
- **Fusidic Acid** : is a bacteriostatic natural antibiotic . It inhibits protein synthesis by interfering with the function of the **elongation factor EF-G**

Inhibitors of Nucleic Acid Synthesis

- Nucleic acids are **blueprints** for the reproduction of every cell; these are DNA and RNA.
- disrupting the formation of nucleic acid results in the destruction of the pathogen
- the ideal antibiotics that target nucleic acid synthesis should be **selective** because the antibiotic might interfere with the host's DNA and RNA
- Antimicrobials in this group are categorized as
 - **Attacking Metabolites**
 - **Attacking processes of synthesis**

Quinolones: highly effective broad spectrum antibiotics that targets, DNA gyrase and DNA topoisomerase IV, both belonging to the type II topoisomerases

✚ Which are required for **DNA replication**, **recombination**, and **repair**.

✚ **gram-negative bacteria** DNA **gyrase-A** subunit is the primary target

✚ **gram-positive bacteria** **topoisomerase IV** is the primary target

✚ The drug follows one of two mechanisms

- ✓ either inhibit the catalytic activity of the enzyme by binding to the active site
- ✓ stabilize the covalent enzyme-DNA complex that is formed during the reaction thus interrupting the religation step .

✚ The **fluoroquinolones** are highly active and bactericidal against a wide range of aerobes and facultative anaerobes.

- ✓ Nalidixic acid and Ciprofloxacin

Folate Inhibitors: are antimetabolites that compete with *p*-aminobenzoic acid

- Folic acid enzymes are necessary for the synthesis of Amino acids
- Folic acid is derived from *para*-aminobenzoic acid (PABA), glutamate, and a pteridine unit.
- Agents that interfere with synthesis of folic acid have selective toxicity because mammalian cells are unable to accomplish this feat and use preformed folate
- Folic acid is an essential coenzyme for the transport of one-carbon compounds in the synthesis of:
 - Purines,
 - Thymidine,
 - Some amino acids, and
 - Indirectly of nucleic acids and proteins.
- The major inhibitors of the folate pathway are the **sulfonamides**, **trimethoprim**, ***para*-aminosalicylic acid**, and the **sulfones**.

Sulfa Drugs

- Includes the sulfonamides, function by inhibiting folic acid biosynthesis in bacteria.
- the structure of sulfonamide resembles that of para aminobenzoic acid (PABA)
- Sulfonamide antibiotics function by inhibiting the enzyme **dihydropteroate synthase (DHPS)** which combines **PABA** and **pteridine** in the initial stage of folate synthesis (inhibiting the formation of DHF)
- Because of the similarity in structure, PABA and sulfonamides compete for binding to the active site of the enzyme.
- Another target of antimetabolites is the enzyme ***Dihydrofolate Reductase*** (DHFR) which **catalyzes a later step** in the pathway in which DHF is reduced to THF
 - This inhibition blocks the formation of **thymidine**, some **purines**, **methionine**, and **glycine**.
 - So the bactericidal effect of Sulfonamide antibiotics is known as “**thymine-less death**.”
- Are broad spectrum bacteriostatic antibiotics
- Sulfonamides are effective against a broad range of gram-positive and gram-negative organisms, such as *Nocardia*, *Chlamydia*, and some protozoa.

Rifampin

- binds to **DNA dependent RNA polymerase** the enzyme that catalyzes the process of **transcription** and inhibits the **initiation of RNA synthesis**.
- binds to the **β -subunit** of RNA polymerase within the DNA/RNA channel.
- inhibits RNA synthesis by directly **blocking the path of the elongating RNA** when the transcript becomes 2 to 3 nucleotides in length
- bactericidal antibiotics that inhibit both gram-negative and gram-positive bacteria including obligate intracellular bacteria.
- One advantage of rifampins as antibiotics is that they can penetrate into mammalian tissues and also into cells.
 - Effective in the treatment of mycobacterial diseases, such as tuberculosis and leprosy
 - But bacteria develop resistance to the drug very quickly.

Table 1. Properties of Some Common Antibacterial Drugs

Drug	Primary Effect	Spectrum
Ampicillin	Cidal	Broad (gram +, some -)
Bacitracin	Cidal	Narrow (gram +)
Carbenicillin	Cidal	Broad (gram +, many -)
Cephalosporins	Cidal	Broad (gram +, some -)
Chloramphenicol	Static	Broad (gram +, -; rickettsia and chlamydia)
Ciprofloxacin	Cidal	Broad (gram +, -)
Clindamycin	Static	Narrow (gram +, anaerobes)
Dapsone	Static	Narrow (mycobacteria)
Erythromycin	Static	Narrow (gram +, mycoplasma)
Gentamicin	Cidal	Narrow (gram -)
Isoniazid	Static or cidal	Narrow (mycobacteria)
Methicillin	Cidal	Narrow (gram +)
Penicillin	Cidal	Narrow (gram +)
Polymyxin B	Cidal	Narrow (gram -)
Rifampin	Static	Broad (gram +, mycobacteria)
Streptomycin	Cidal	Broad (gram +, -; mycobacteria)
Sulfonamides	Static	Broad (gram +, -)
Tetracyclines	Static	Broad (gram +, -; rickettsia and chlamydia)
Trimethoprim	Cidal	Broad (gram +, -)
Vancomycin	Cidal	Narrow (gram +)

Mechanisms of Antibacterial Drug Action

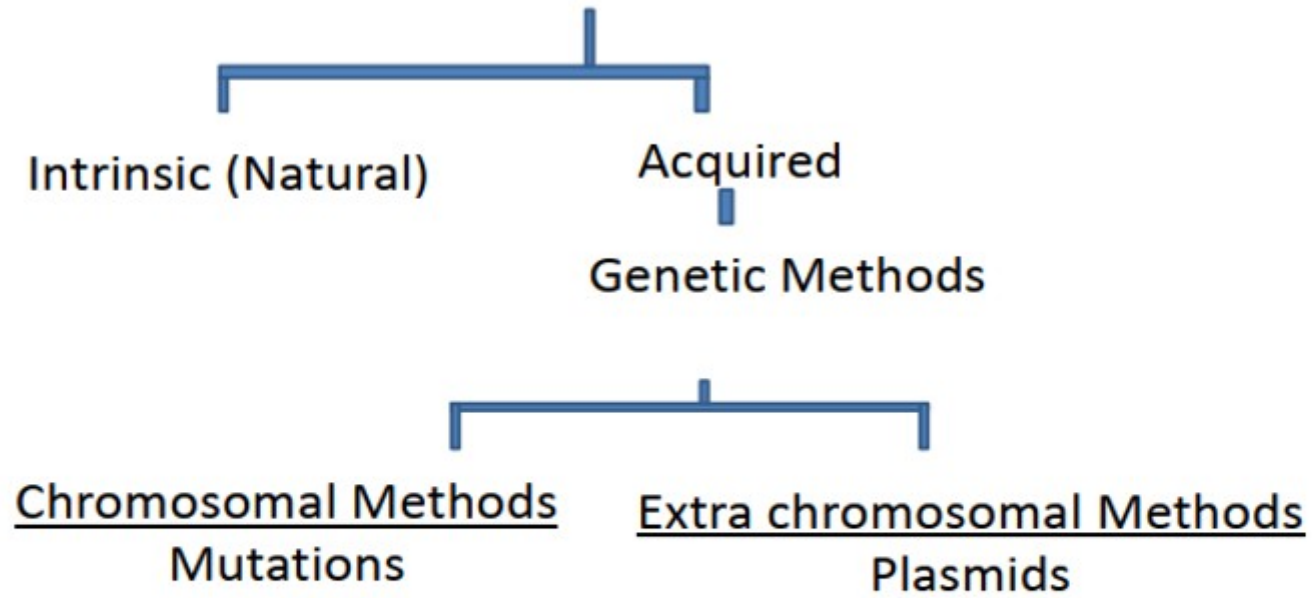
Drug	Mechanism of Action
Cell Wall Synthesis Inhibition	
Penicillin	Inhibit transpeptidation enzymes involved in the cross-linking of the polysaccharide chains of the bacterial cell wall peptidoglycan. Activate cell wall lytic enzymes.
Ampicillin	
Carbenicillin	
Methicillin	
Cephalosporins	
Vancomycin	Binds directly to the D-Ala-D-Ala terminus and inhibits transpeptidation.
Bacitracin	Inhibits cell wall synthesis by interfering with action of the lipid carrier that transports wall precursors across the plasma membrane.
Protein Synthesis Inhibition	
Streptomycin	Binds with the 30S subunit of the bacterial ribosome to inhibit protein synthesis and causes misreading of mRNA.
Gentamicin	
Chloramphenicol	Binds to the 50S ribosomal subunit and blocks peptide bond formation through inhibition of peptidyl transferase.
Tetracyclines	Bind to the 30S ribosomal subunit and interfere with aminoacyl-tRNA binding.
Erythromycin and clindamycin	Bind to the 50S ribosomal subunit and inhibit peptide chain elongation.
Fusidic acid	Binds to EF-G and blocks translocation.
Nucleic Acid Synthesis Inhibition	
Ciprofloxacin and other quinolones	Inhibit bacterial DNA gyrase and thus interfere with DNA replication, transcription, and other activities involving DNA.
Rifampin	Blocks RNA synthesis by binding to and inhibiting the DNA-dependent RNA polymerase.
Cell Membrane Disruption	
Polymyxin B	Binds to the plasma membrane and disrupts its structure and permeability properties.
Metabolic Antagonism	
Sulfonamides	Inhibit folic acid synthesis by competition with <i>p</i> -aminobenzoic acid.
Trimethoprim	Blocks tetrahydrofolate synthesis through inhibition of the enzyme dihydrofolate reductase.
Dapsone	Interferes with folic acid synthesis.
Isoniazid	May disrupt pyridoxal or NAD metabolism and functioning. Inhibits the synthesis of the mycolic acid "cord factor."

Antibiotic Resistance

- Defined as **micro-organisms** that are **not inhibited** by usually achievable **systemic concentration** of an **antimicrobial** agent with normal dosage schedule and / or fall in the minimum inhibitory concentration (MIC) range **OR**
- Loss of susceptibility of bacteria to the killing (bacteriocidal) or growth-inhibiting (bacteriostatic) properties of an antibiotic agent.



Genetics of Resistance



Classification of Antibiotic Resistance

- Antibiotic resistance in bacteria can be of two types:
 - Intrinsic resistance
 - Acquired resistance
- **Intrinsic resistance:** is when by **virtue** of their structural or functional features, some bacteria are **naturally** resistant to some antibiotics without having prior exposure to the antibiotics.
 - **Lack target:**
 - **Innate efflux pumps:**
 - **Drug inactivation:**
 - gram-negative bacteria are intrinsically resistant to **vancomycin**
 - Aerobic bacteria are intrinsically resistant to **metronidazole** , which requires an anaerobic environment to be reduced to its active form

Acquired resistance:

- In a population of antibiotic sensitive bacteria, some cells may acquire the ability to be resistant to the antibiotic.
- Thus, unlike intrinsic resistance , which is effective in all cells of a certain species, acquired resistance can be observed in **only a subpopulation** of any bacterial species.
- Acquired resistance development can take place by two different mechanisms:
 - By point mutations and
 - By resistance gene acquisition.

Resistance Development by Point Mutations

- Two types: natural methods and induced methods

- **Natural methods or Replication errors**

- random mutations, also known as **spontaneous** mutations

- **Happens due to error during rapid replication of the bacteria**

- **Induced methods**

- harsh environmental conditions(strong ionizing radiations or oxidizing or alkylating chemicals)

- not relevant for resistance development but

- sub lethal concentration of antibiotic can cause point mutations

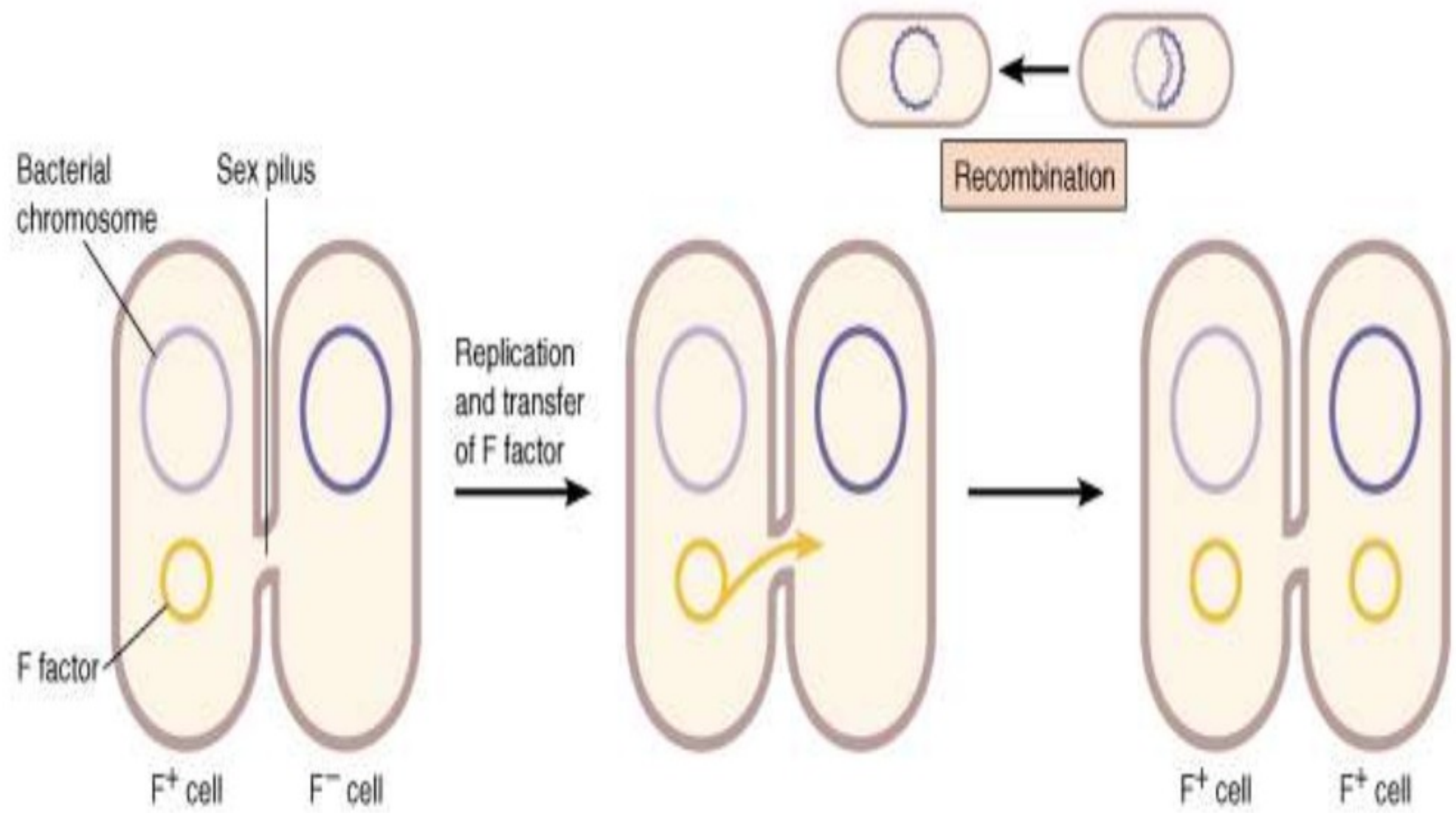
Resistance Development by Resistance Gene Acquisition

- Most common mechanism by which resistance develops
- It is facilitated by the fact that bacteria are **prokaryotic** and by the presence of small pieces of **DNA** called **plasmids** that exist in a bacterial cell separate from the chromosome.
- Thus, the genetic material of bacteria is free-floating within the cell, making it open to **gene** transfer
- Antibiotic resistance genes usually reside in either the **chromosome** or **plasmids** or **transposons** or **integrans**.
- Transfer of resistance genes between bacteria can take place in three ways:
 - Bacterial Conjugation
 - Bacterial Transformation
 - Transduction/Transfection

CONJUGATION

- It is the process of **gene transfer** via **sex pili** between cells of opposite mating types that are in **physical contact** (cell-to-cell) with each other.
- Requires the presence of a special plasmid called the **F plasmid**.
- Bacteria that have a F plasmid are referred to as F⁺ or male. Those that do not have an F plasmid are F⁻ or female.
- The F plasmid consists of 25 genes that mostly code for production of sex pilli.
- A conjugation event occurs when the male cell extends his sex pilli and one attaches to the female.
- This attached pilus is a temporary cytoplasmic bridge through which a replicating F plasmid is transferred from the male to the female.
- When transfer is complete, the result is two male cells.

- In order to be transferred, one strand of the plasmid DNA is cut at a site called the **origin of transfer (oriT)**
- The cut single strand is transferred and then joined to make it circular
- The second strand is synthesized in both the donor and recipient strain.
- by the method of bacterial conjugation , more and more antibiotic sensitive bacteria can acquire antibiotic resistance genes and become resistant.
- Main mechanism for spread of resistance



Transduction/Transfection.

- This is the process by which DNA is transferred between bacteria using **bacteriophages** as intermediaries.
- Transduction does not require physical contact between the cell donating the DNA and the cell receiving the DNA (which occurs in conjugation)
- Bacteriophages are viruses that infect bacteria.
- When a bacteriophage lyses a bacterium, the bacterial genome is fragmented into small parts.
- During viral replication, a small part of the bacterial genome is sometimes mistakenly incorporated into the viral genome and **is carried into recipient cell during infection.**
- In the process of multiplication the bacteriophage can incorporate some of the bacterial DNA into their DNA and can then transfer the DNA to other bacteria that they infect.
- antibiotic resistance genes are transferred between bacteria thus contributing to spread of resistance.

➤ The viral transduction process can be categorized into two forms:

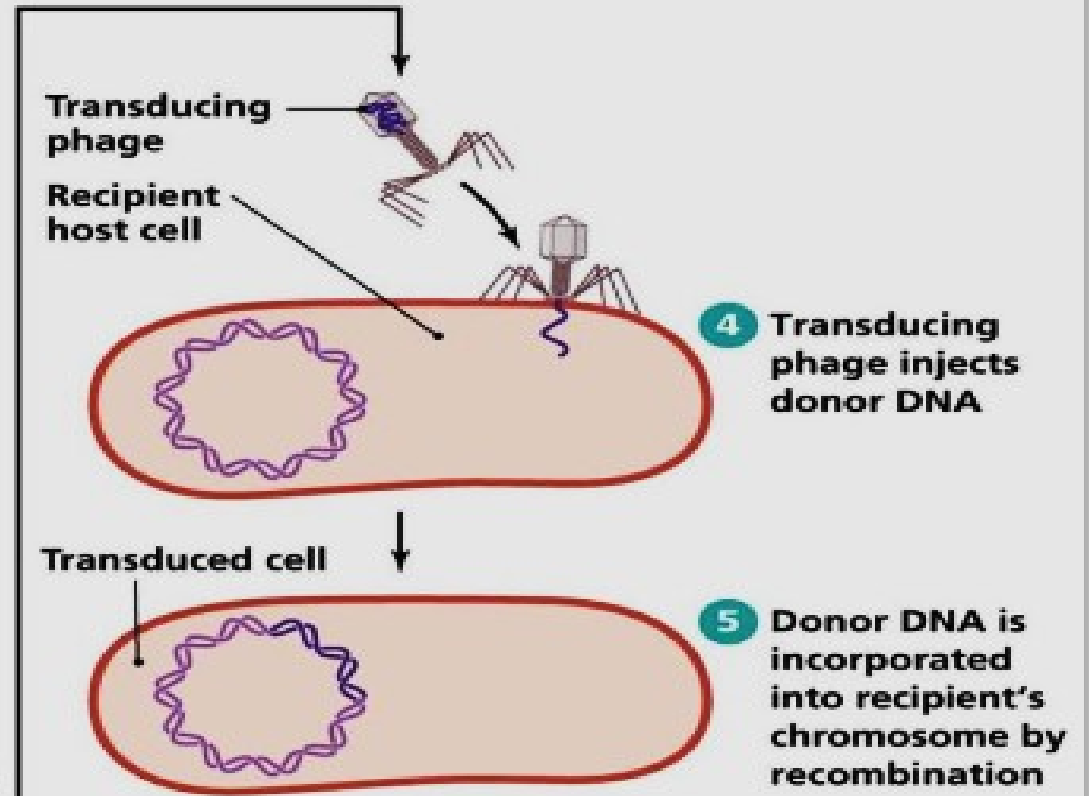
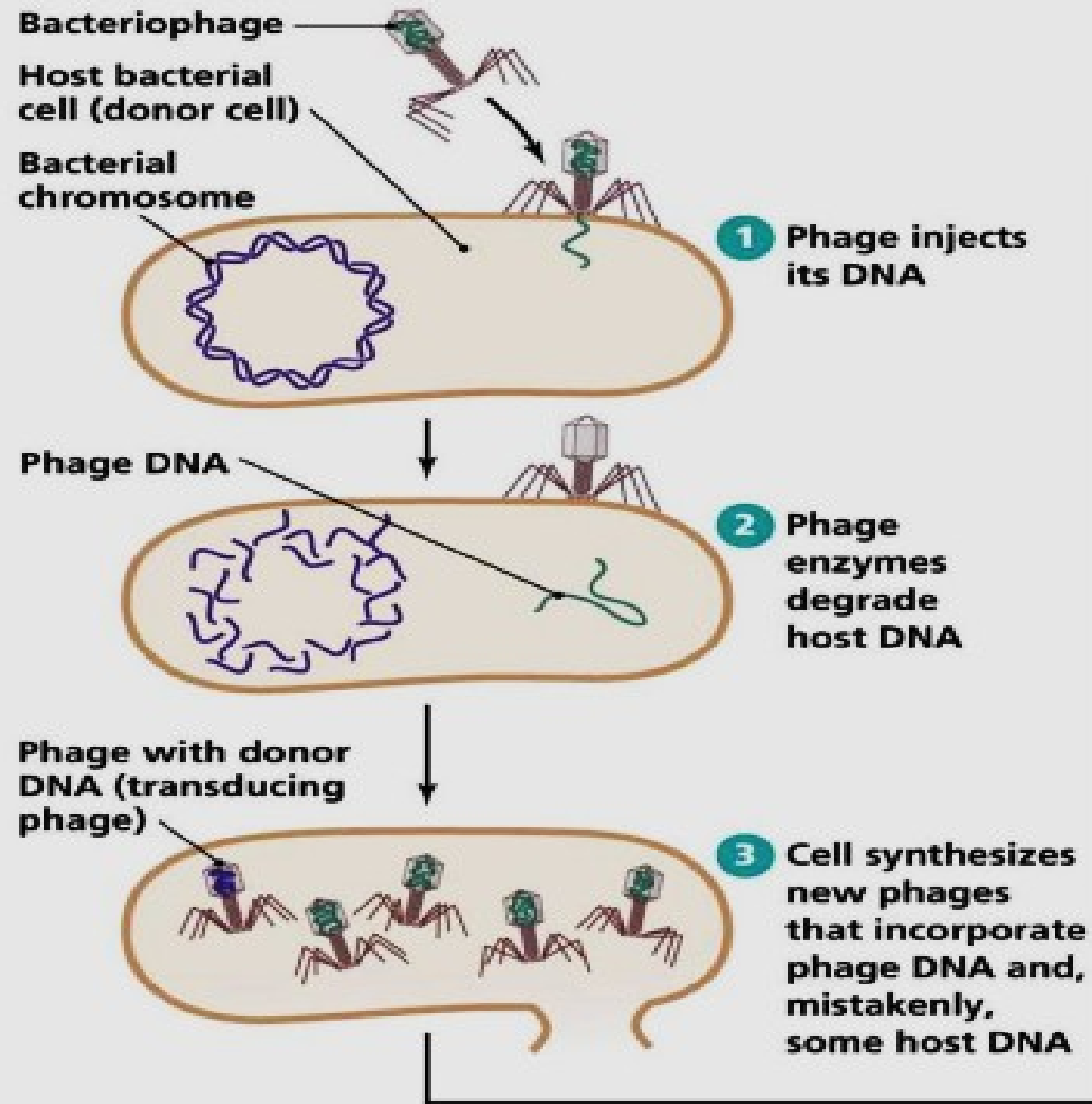
Generalized: any part of the bacterial genome is transferred to the new host

Specialized: only specific parts of the bacterial genome are transferred during specialized transduction

➤ Transduction has an advantage over conjugation in that which doesn't require physical contact and resistant to the DNase enzyme

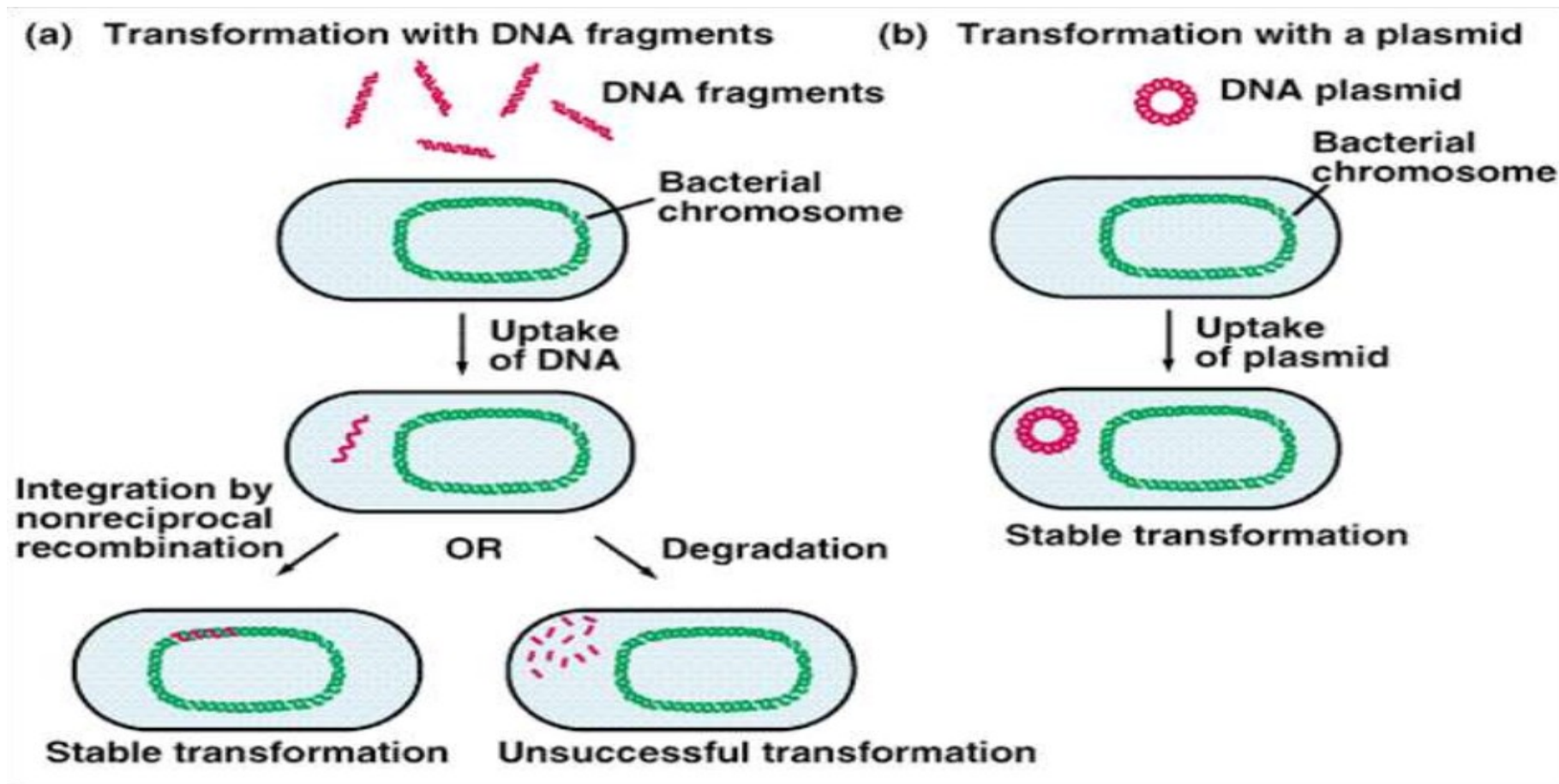
- Because of the high specificity of bacteriophages, transduction is typically limited to bacteria of the same species.

➤ Transduction happens through either the lytic cycle or the lysogenic cycle.



Transformation

- This is the process by which bacteria take in DNA from outside which is usually released from dead bacteria.
- This process doesn't require a living donor cell and only requires free DNA in the environment.
- Transformation is adopted as the most common method of gene transfer as it is the **best way** for the transfer of **artificially altered DNA** into recipient cells.
- The success of transformation depends on the competence of the host cell.
- Competence is the ability of a cell to incorporate naked DNA in the process of transformation
- Several bacteria, including *Escherichia coli*, can be artificially treated in the laboratory to increase their transformability by chemicals
- In most of the naturally competent bacteria, the free DNA binds to the bacteria, and the DNA is integrated into the chromosomal DNA.



There are four steps in transformation:

1. development of competence
2. binding of DNA to the cell surface
3. processing and uptake of free DNA
4. integration of the DNA into the chromosome by recombination

Common Mechanisms of Resistance

The major mechanisms of bacterial resistance are:

1. Decreased Antibiotic Penetration and Efflux
2. Alterations of an antimicrobial target,
3. inactivation of an antimicrobial by an enzyme produced by the microorganism

Decreased permeability of cell wall

- Some bacteria develop resistance to antibiotic by changing their permeability to the drug
 - Effective concentration antibiotics not achieved inside the bacterial cell
 - This mainly due to modification of their **porin protein**
 - *a shift in the type of porins expressed , a change in the level of porin expression and impairment of the porin function*
- Vancomycin a glycopeptide antibiotic in GN bacteria
- *P. aeruginosa* to tetracyclines
- Streptococci to aminoglycosides

Efflux Pumps

- 🍎 Are transport proteins involved in the extrusion of toxic substrates (including virtually all classes of clinically relevant antibiotics)
- 🍎 These proteins are found in both Gram-positive and -negative bacteria as well as in eukaryotic organisms.
- 🍎 Pumps may be specific for one substrate or may transport a range of structurally dissimilar compounds
- 🍎 In the prokaryotic kingdom there are five major families of efflux transporter
 - MF (major facilitator),
 - MATE (multidrug and toxic efflux),
 - RND (resistance-nodulation-division),
 - SMR (small multidrug resistance) and
 - ABC (ATP binding cassette)
- 🍎 These families differ in terms:
 - Structural conformation,
 - Energy source,
 - Range of substrates they are able to extrude and
 - In the type of bacterial organisms in which they are distributed
 - Tetracycline resistance is one of the classic examples

Changes in Target Sites

- A common strategy for bacteria to develop antimicrobial resistance is to avoid the action of the antibiotic by interfering with their target site.
- Antimicrobics act by binding and inactivating their target, which is typically a crucial enzyme or ribosomal site
- If the target is altered in a way that decreases its affinity for the antimicrobial the inhibitory effect will be proportionately decreased
- To achieve this, bacteria have evolved different tactics
 - protection of the target (avoiding the antibiotic to reach its binding site

alters the ribosomal conformation, preventing rebinding of the antibiotic

- modifications of the target site

the most common mechanisms of antibiotic resistance and consist of

-  Point mutations in the genes encoding the target site
-  enzymatic alterations of the binding site (e.g. addition of methyl groups)
-  replacement or bypass of the original target

Inactivation of an antimicrobial by an enzyme

- Is the most powerful and robust of the resistance mechanisms.
- Enzymes produced by resistant bacteria may inactivate the antimicrobial in the cell, in the periplasmic space, or outside the cell.
- They may act on the antimicrobial molecule by disrupting its structure or by catalyzing a reaction that chemically modifies it.

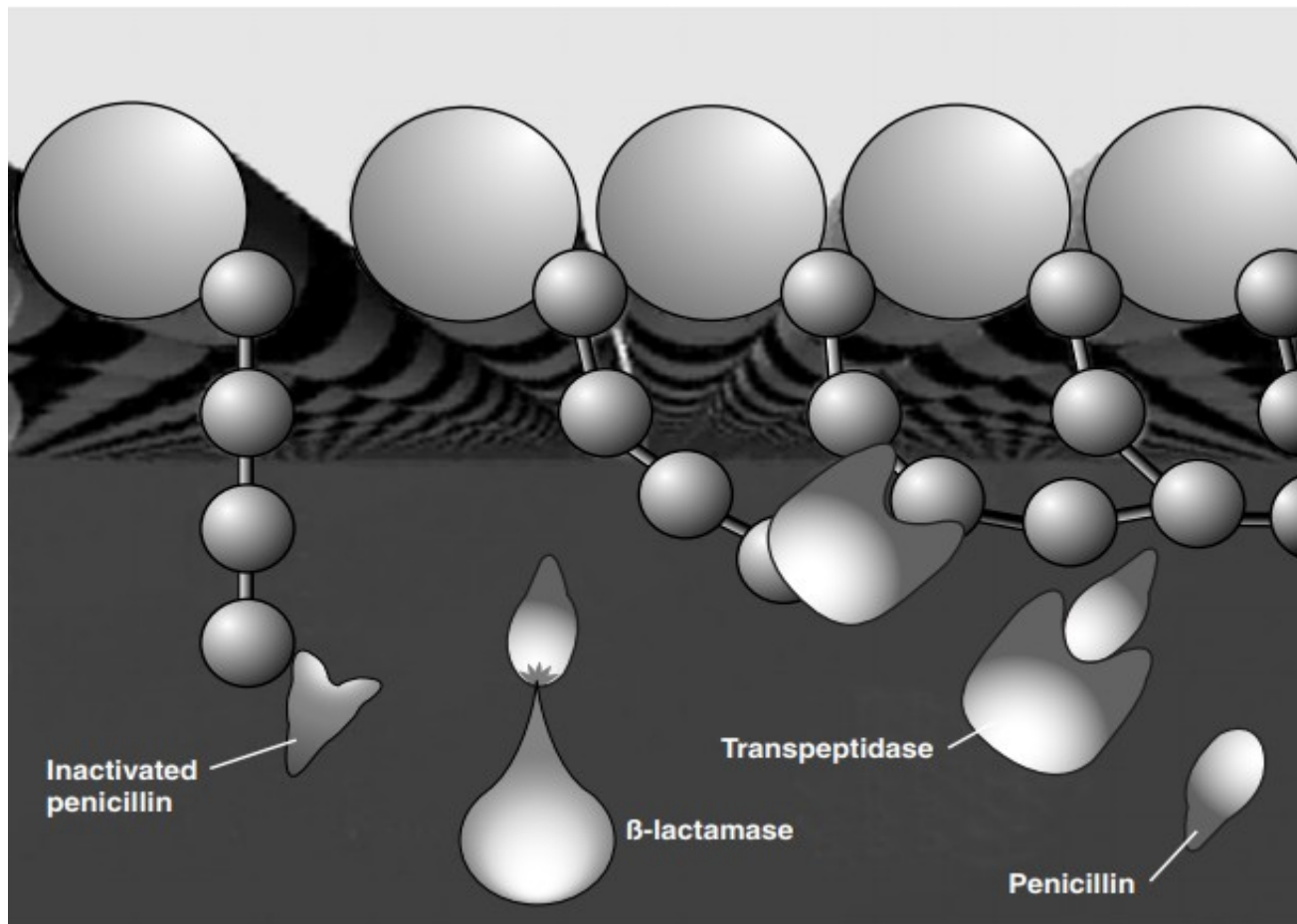
Chemical alterations of the antibiotic

- The production of enzymes capable of introducing chemical changes to the antimicrobial molecule is a well-known mechanism of acquired antibiotic resistance in both gram-negative and gram-positive bacteria
- Most of the antibiotics affected by these enzymatic modifications exert their mechanism of action by inhibiting protein synthesis at the ribosome level

- Many types of modifying enzymes have been described, and the most frequent biochemical reactions they catalyze include:
 - ✓ **Acetylation**(aminoglycosides, chloramphenicol, streptogramins)
 - ✓ **Phosphorylation** (aminoglycosides, chloramphenicol)
 - ✓ **Adenylation** (aminoglycosides, lincosamides)
- The resulting effect is often related to steric hindrance that **decreases the avidity** of the drug for its target
- One of the best examples of resistance via modification of the drug is the presence of aminoglycoside modifying enzymes (AMEs)
 - covalently modify the hydroxyl or amino groups of the aminoglycoside molecule.

Destruction of the antibiotic molecule

- The main mechanism of β -lactam resistance by the action of β -lactamases
- Beta-Lactamase is a general term referring to any one of hundreds of bacterial enzymes able to break open the beta-lactam ring and inactivate various members of the beta-lactam group.
- The first was discovered when penicillin-resistant strains of *S. aureus* emerged and the enzyme was called **penicillinase**
- In order to overcome this problem, new β -lactam compounds with wider spectrum of activity and less susceptibility to penicillinases (such as ampicillin) were manufactured
- a new plasmid-encoded β -lactamase capable of hydrolyzing ampicillin was found among gram negatives
- Each beta-lactamase is a distinct enzyme with its own physical characteristics and substrate profile.



Enzymatic inactivation resistance.